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Introduction to Data Communications

Topic 01 – Introduction To Networks

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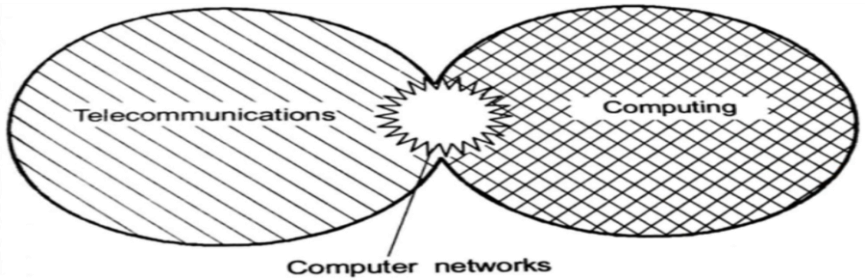


Networking

- What is a network?
- Data network elements
- Transmission Technology
  - -Point-2-point (unicast)
  - -Multi-point (multicast, broadcast)
- Scales of network
  - -LAN, MAN, WAN,
  - -WLAN, WMAN, WWAN, WPAN

What is a network?

- A network is a set of interlinked devices or nodes that can transmit and share resources.
- E.g. The ancient Rome water supply system and home networks.



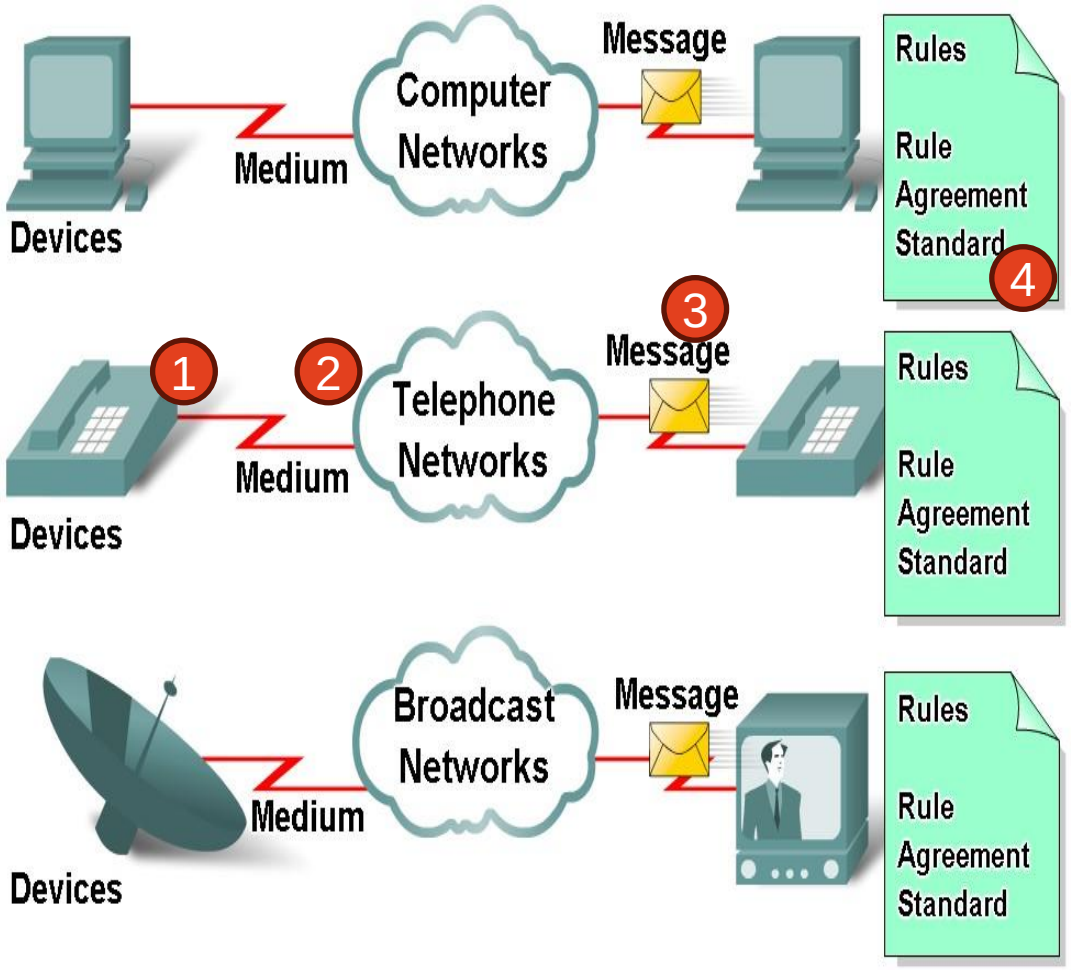
What is a Computer Network?

- A computer network is a set of interlinked computers that are involved in data exchange and collaboration through different ways such as wired or wireless connections.
- It is involved in effective data transmission, remote access, and resource sharing between devices.
- E.g. Internet, Intranet, WAN, LAN , WSN



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Data Networks and Elements



Rules *(protocols)*

- A protocol is a set of rules that control how data is transmitted, received, and processed between devices on a network.
- It controls how messages flow across the network.
- It controls how communication between peers will take place.

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Transmission Technology

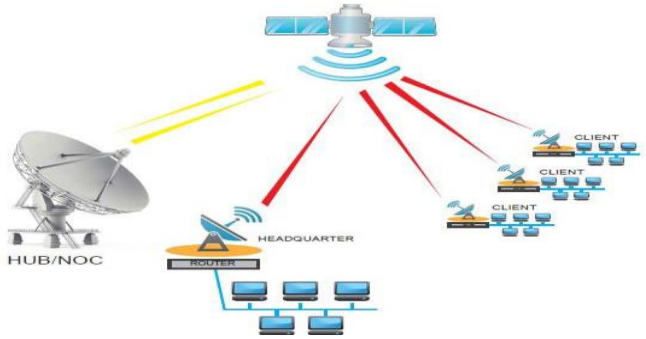
Point-to-point Network

- There is only one sender and one receiver involved in the process and they have dedicated links.
- Unicasting
- It is a type of network topology where two devices are directly connected through a dedicated communication link.
- Data flow happens only in one direction



Multipoint (Multidrop) Network

- The link capacity is used by multiple nodes.
- Broadcasting, Multicasting
- A multipoint network, is also called a multidrop network.
- It is a network topology where many devices are linked to a single communication channel or link.
- E.g. Bus Topology



Scales of Network

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

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Local Area Network (LAN)

- A LAN, is a network that links computers and other devices.
- It works in a limited geographical area
- E.gh. home, office, or school campus.
- In a LAN, devices are usually linked with Ethernet cables or wireless connections.
- This allows them to communicate and share resources such as files, printers, and internet access.

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- LAN is a privately owned network.
- It is restricted in size usually within a building or campus.
- Transmission technology a single cable – 10Mbps/100Mbps/10Gbps(400 Gbps in 2017/2018)
- Wireless transmission
- 2Mbps/11Mbps/54Mbps/108Mbps/250Mbps/1000Mbps
- Various topologies are possible- Bus, Ring and Star

Metropolitan Area Network (MAN)

- It is a large LAN and it uses the same LAN technology.
- A Metropolitan Area Network, covers a city area, or a big campus. A MAN usually covers a bigger geographical area than the LAN however it is a WAN (Wide Area Network).
- Size- It covers a group of nearby offices or a city.

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Metropolitan Area Network (MAN)

Transmission technology of MAN:

- High-speed backbone linking multiple LANs
- Digital Subscriber Line (DSL)
- TV cables
- Fiber Distributed Data Interface (FDDI)
- Distributed-Queue Dual-Bus (DQDB) network

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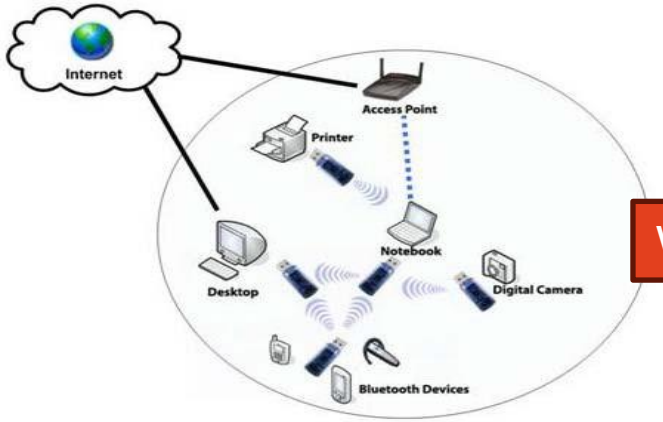
Wide Area Network (WAN)

- A Wide Area Network, covers a bigger geographical area.
- It links many LANs and other networks over long distances.
- WANs can cover cities, countries, or even continents, letting organizations connect geographically dispersed locations.
- Large MAN uses different technologies.

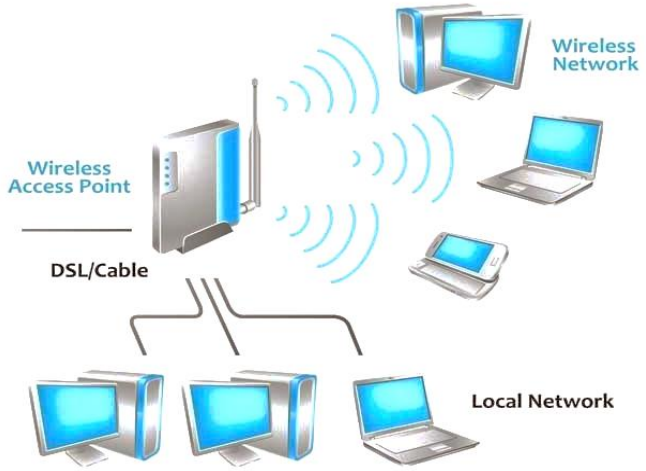
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Wireless Networks

- Wireless Personal Area Networks (WPAN)
- Wireless LAN (WLAN)
- Wireless MAN (WMAN)
- Wireless WAN (WWAN)
- Cellular Networks



WPAN



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Classification Of Networks

- Circuit-switched networks
- Packet-switched networks
  - -Datagram networks
  - -Virtual Circuit networks

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Circuit-Switched Networks

Circuit-Switched Networks:

- They are a kind of telecommunications network.
- It has a reserved communication path, or circuit which is established between two nodes for the duration of a connection.
  - Circuit Establishment
  - Data Transfer
  - Circuit Termination

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Packet-Switched Networks

- It is a telecommunications network where data is transmitted in separate units called packets.
  - Datagram (Packet Switched) Networks
  - Virtual Circuit (Packet Switched) Networks

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Datagram PS Network

Datagram Packet Switched Network:

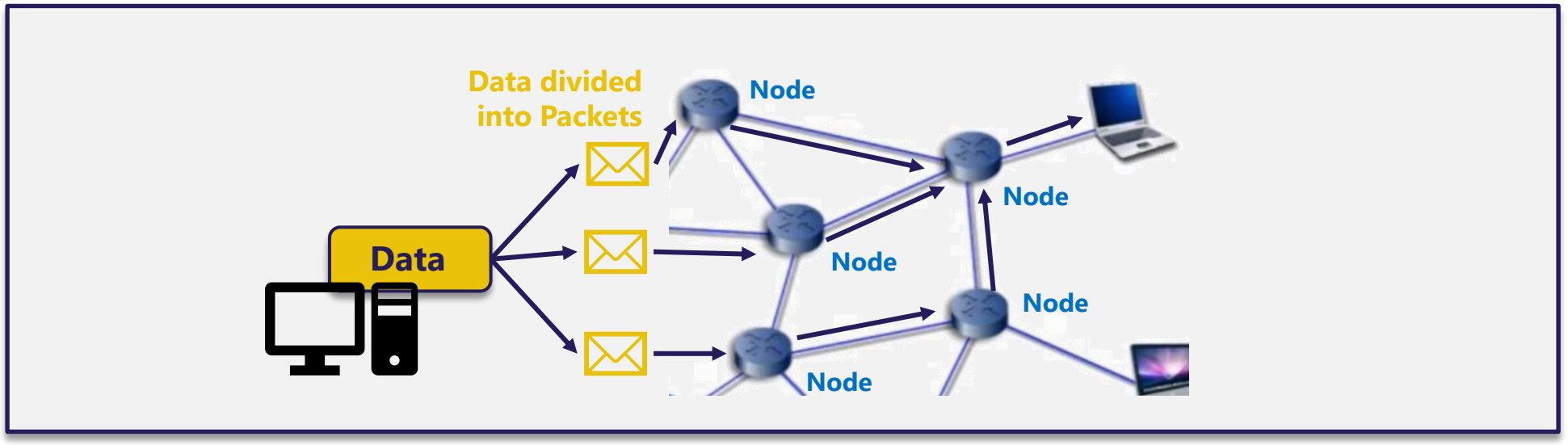
Data is sent in the form of individual packets, also called datagrams.

It is done without the need for establishing a pre-defined link or virtual circuit between sender and receiver.

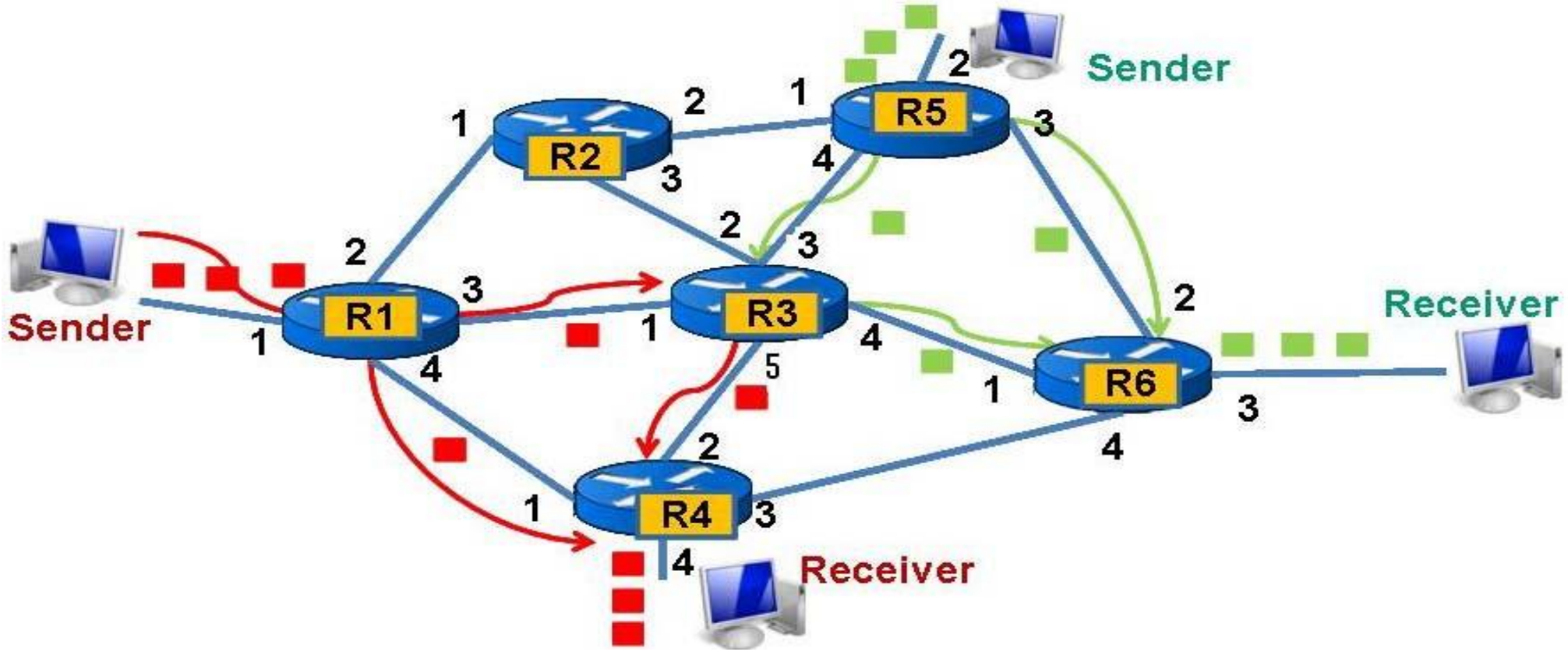
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Datagram PS Network

- At a Node, packets are saved temporarily and then forwarded to the next node (store-and-forward)
- No physical capacity is allocated for packets



Datagram PS Network



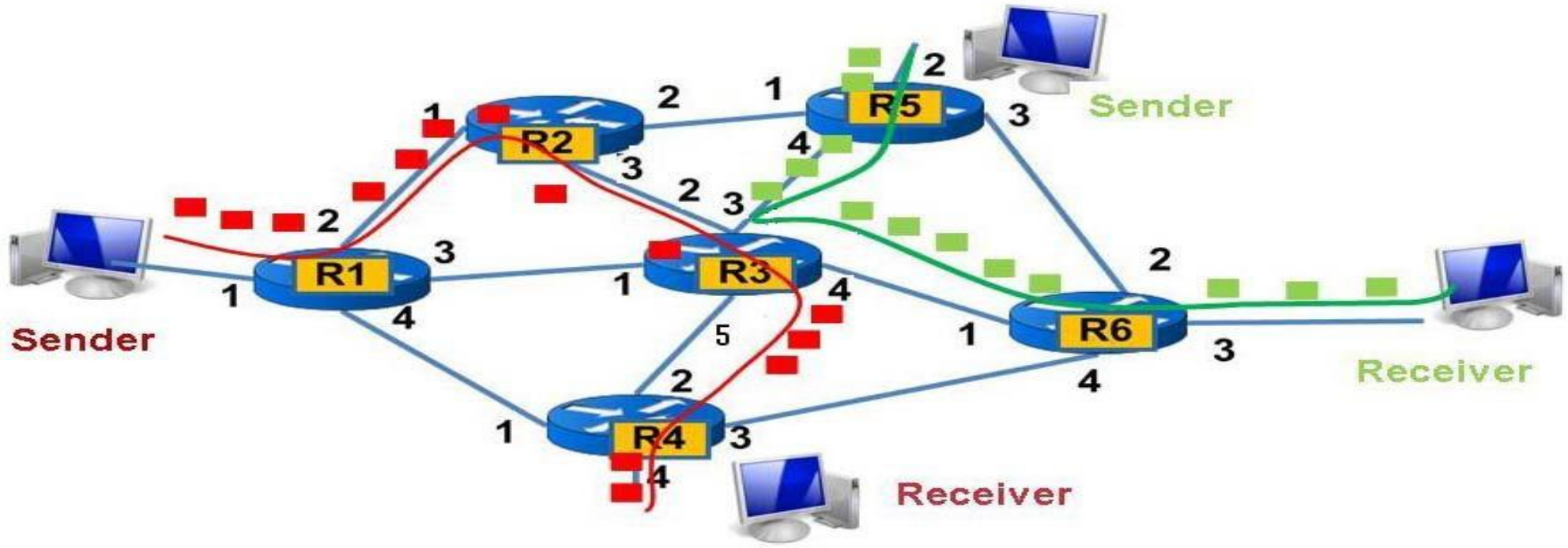
Virtual Circuit PS Network

Hybrid:

- Packet Switching + Circuit Switching
- A Virtual Circuit (VC) in a packet-switched network is a logical transmission path that is created between two network nodes to enable data transmission
- A preplanned route (between stations)
- All packets will take the same route for the duration of the logical connections
- But there is no dedicated path between the two stations

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Virtual Circuit PS Network



Circuit Switching	Datagram Packet Switching	Virtual Circuit Packet Switching
Dedicated transmission path	No dedicated transmission path	No dedicated transmission path
Continuous transmission	Transmission of packets	Transmission of packets
Path stays fixed for entire connection	Route of each packet is independent	Path stays fixed for entire connection
Call setup delay	No setup delay	Call setup delay
No queuing delay	Queuing delay at switches	Queuing delay at switches
Busy signal overloaded network	Delay increases in overloaded networks	Delay increases in overloaded networks
Fixed bandwidth for each circuit	Bandwidth shared by all packets	Bandwidth shared by all packets
No overhead after call setup	Overhead in each packet	Overhead in each packet

Common Networking Technologies

- Internet
- ATM

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Common Networking Technologies



Public Switched Telephone Networks (PSTN)

- It is the biggest worldwide computer network, specializing in voice
- Switching: Circuit Switching

The Internet

- A newer global and public information infrastructure
- Switching: Datagram Packet Switching (Mostly)



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Common Networking Technologies

ATM (asynchronous transfer mode; obsolete)

- Flexibility and low-cost ~= Internet
- End-to-end service guarantees ~= telephone network
- To replace telephone networks and data networks
- Switching: VC Packet Switching
- Inefficient due to overheading (48-byte cell with 5-byte header)



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INTERNET

Internet

- A network of geographically spread out computers
- A group of interlinked networks is called internetwork or the internet
  - One of the common forms of the internet is a set of LANs connected by a WAN
- The end users use the Internet via Internet Service providers (ISPs)
- History
  - -Early 1980: First personal computer
  - -Early 1980: Commercial use of the Internet
  - -1991: World Wide Web (www)



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Layers, Services and Protocols

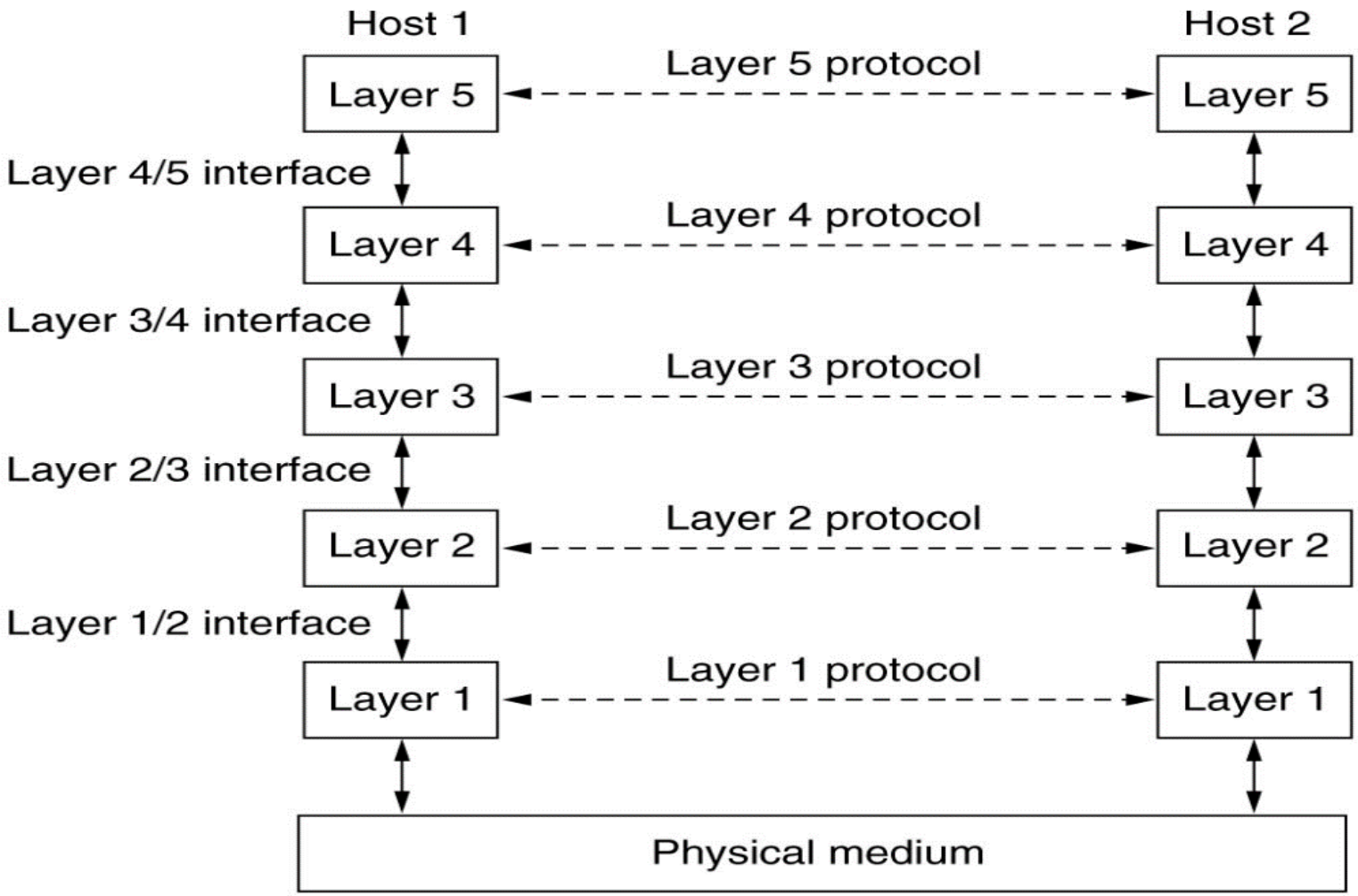
- Layering? Services? Protocol?
- Intro to OSI-7-layer model
- Intro to TCP/IP 5-layer model
- OSI 7-layer model vs TCP/IP 5-layer model

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Why Layering?

- Layering in networking arranges protocols in graded layers, with each layer having specific functions.
- It enables abstraction, independence, and interoperability among network components.
- This modular approach make things easier for development, maintenance, and troubleshooting of network systems.
- Each layer provides a service to the layer above
- Each layer operates according to a protocol
- Protocols can be changed without affecting other layers, higher or lower

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Host 1
Host 2
Layer 5
Layer 5 protocol
Layer 5
Layer 4/5 interface
Layer 4
Layer 4 protocol
Layer 4
Layer 3/4 interface
Layer 3
Layer 3 protocol
Layer 3
Layer 2/3 interface
Layer 2
Layer 2 protocol
Layer 2
Layer 1/2 interface
Layer 1
Layer 1 protocol
Layer 1
Physical medium

Network Protocol?

- Set of rules that manages peer-to-peer communication
- Network Architecture
  - The set of layers of protocols
- Protocol has specific syntax
  - Format of the data block
  - Uses control information for coordination
  - Sequencing
  - Speed matching

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OSI MODEL

- **Open System Interconnect**

  -Introduced by ISO in 1984

  -The OSI model is a theoretical framework that regulates the tasks of a telecommunication or computing system into seven separate layers.

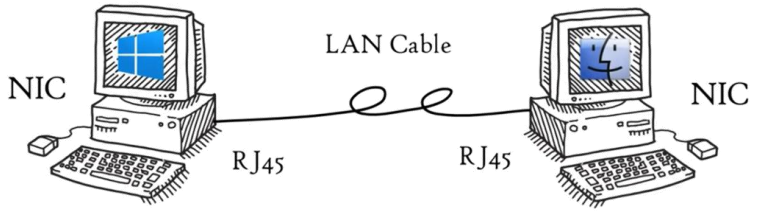
- **Each layer:**

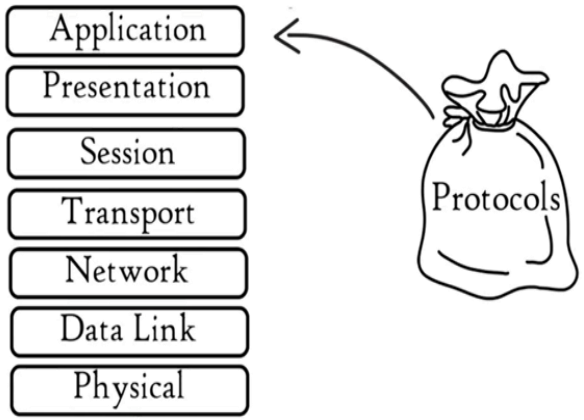
  -Package of protocols

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OSI MODEL

- 7 Layers
  - -**P**lease – Physical
  - -**D**o - Data Link
  - -**N**ot – Network
  - -**T**hrow – Transport
  - -**S**ausage – Session
  - -**P**izza – Presentation
  - -**A**way – Application





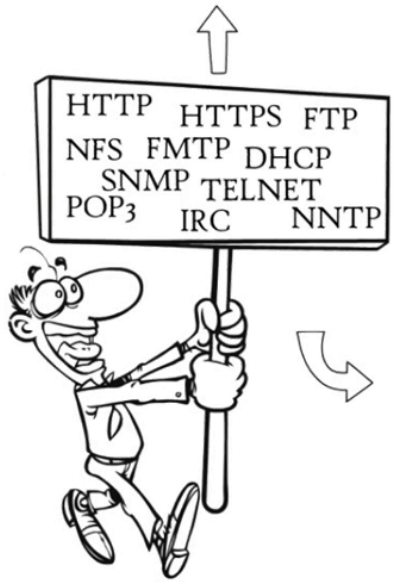
Functions of Each Layer of OSI Model

- **Physical Layer**: It sends raw data bits through the physical medium, specifying electrical, mechanical, and procedural features.
- **Data Link Layer**: It takes care of data framing, error detection, and provides access to the physical medium, making sure of reliable point-to-point and point-to-multipoint communication.
- **Network Layer**: It takes care of routing and delivering packets between different networks, and it also provides logical addressing, routing, and congestion control.

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Transport Layer: It takes care of consistent end-to-end data delivery, performing segmentation, flow control, error recovery, and multiplexing.
Session Layer: Manages session establishment, maintenance, and termination between applications, facilitating dialog control and synchronization.
Presentation Layer: Translates, encrypts, and formats data for compatibility between different systems, handling data compression and encryption.
Application Layer: Provides user interfaces and services for accessing network resources, supporting protocols such as HTTP, FTP, and SMTP.

Application Layer: Network Applications









File Transfer

Web Surfing

Emails

Virtual Terminals











FTP

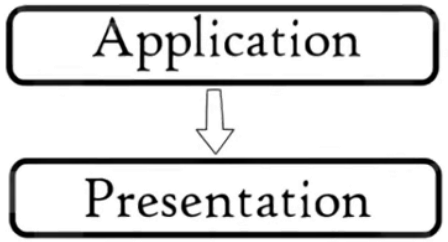
HTTP/S

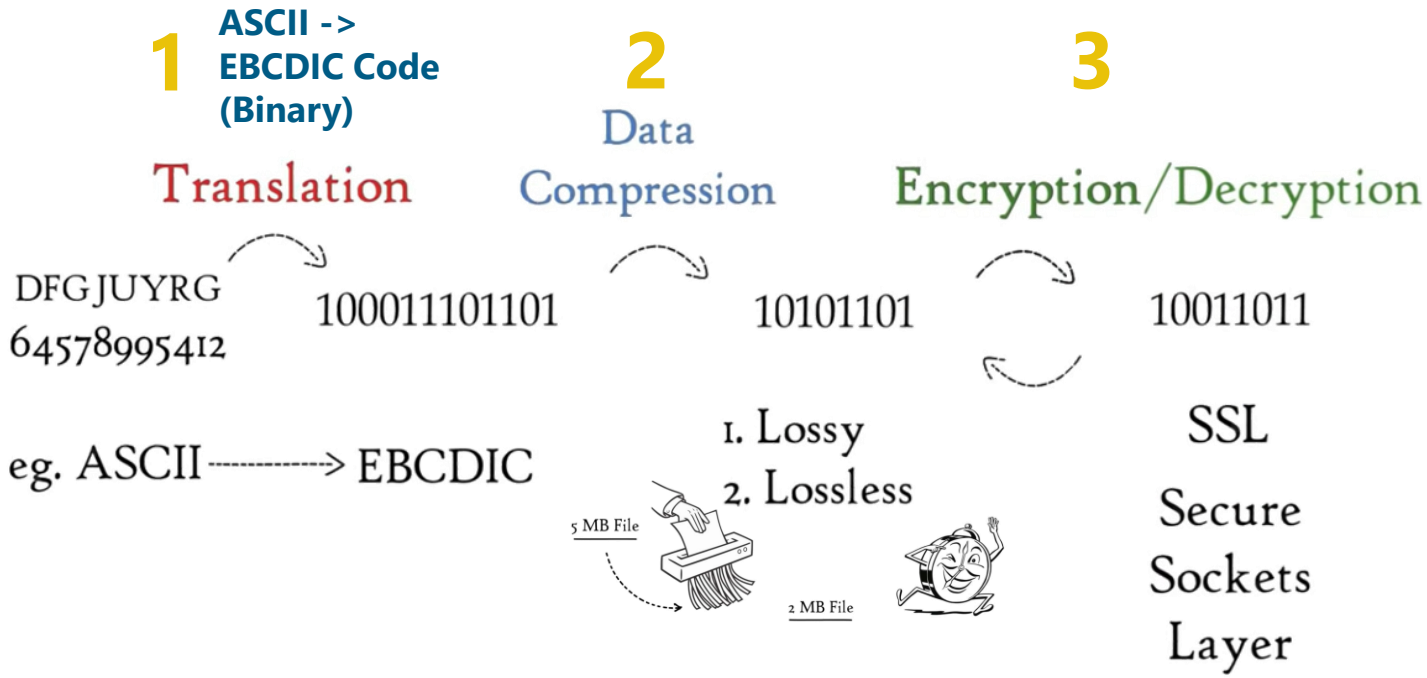
SMTP

Telnet

Presentation Layer

- Used by networks

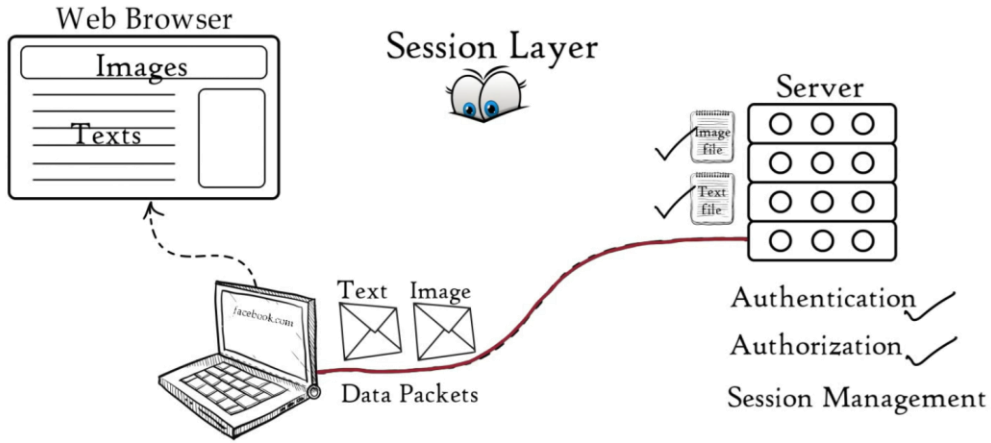




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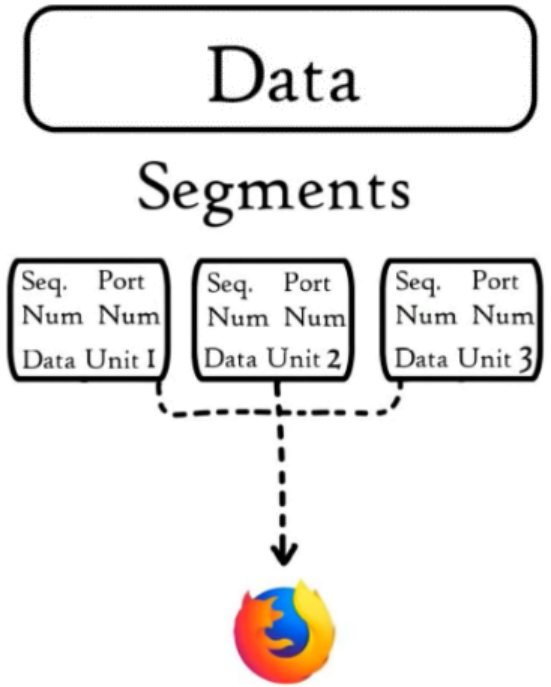
Session Layer

- Setting up and managing connections
- Session Layer Helpers: APIs, NETBIOS
  - Helps applications to communicate with each other
- **Session Management**
- Authentication, Authorization
- **For example,**
  - Web browser performs:
    - Session maintenance across different tabs



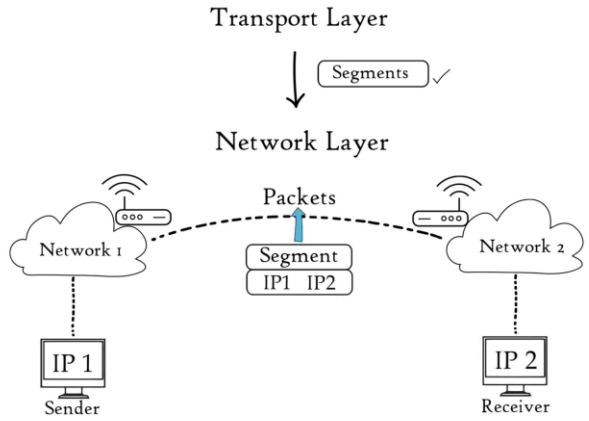
Transport Layer

- This layer sends data from one application in a host to another application in another host
- Segmentation
  - Data -> segments
  - Segment:
    - Port number
    - Seq number
    - Payload (data)



Network Layer

- The network layer plays an important role in aiding communication between devices across interconnected networks.
- It is mainly in charge of routing data packets from the source to the destination, irrespective of the underlying physical network topology.



Data Link Layer

Physical Addressing

- MAC (Media Access Control) Address
- Hardcoded by manufacturer to NIC

Two Sub Layers

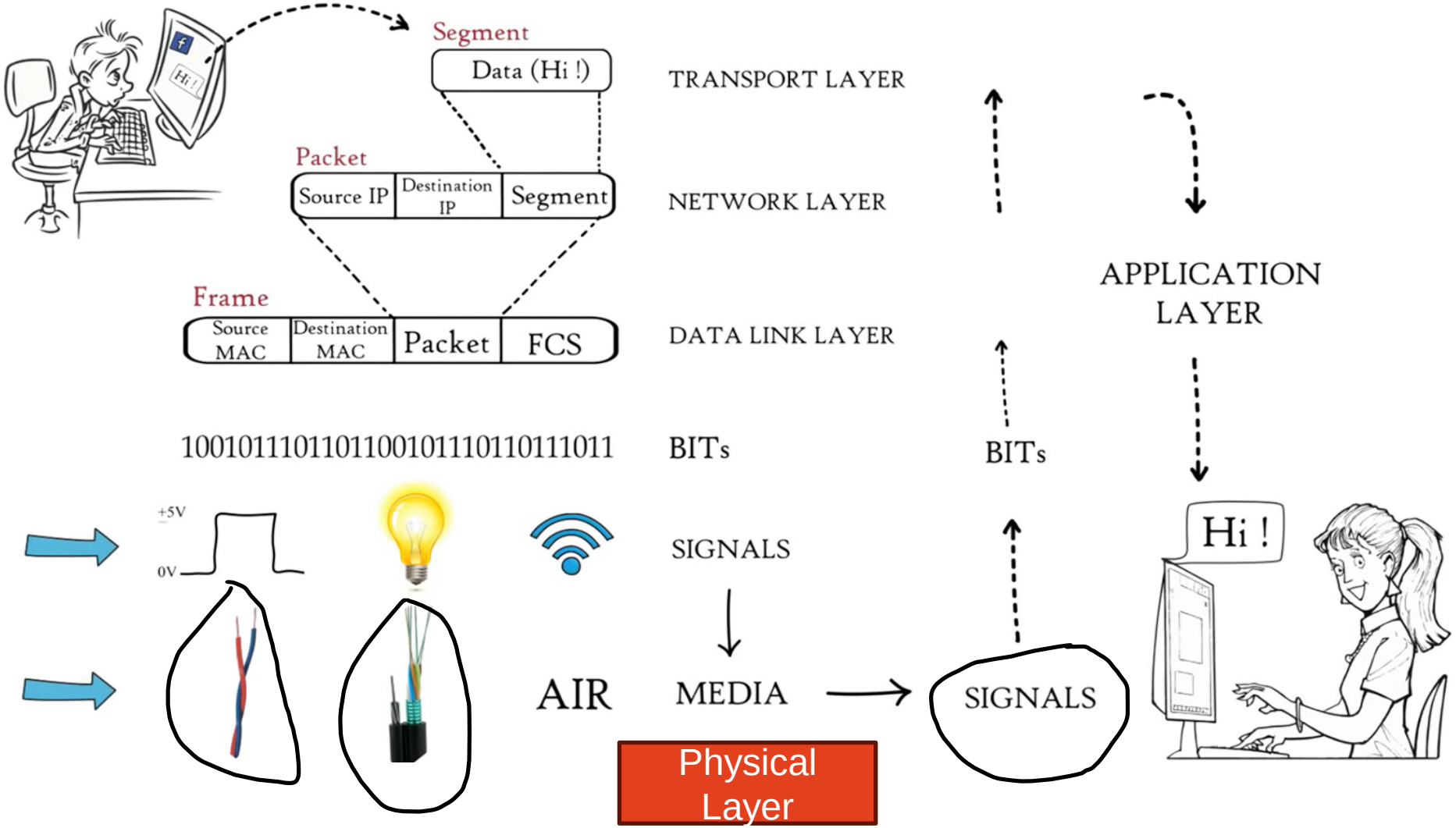
-**Logical Link Control (LLC)**:

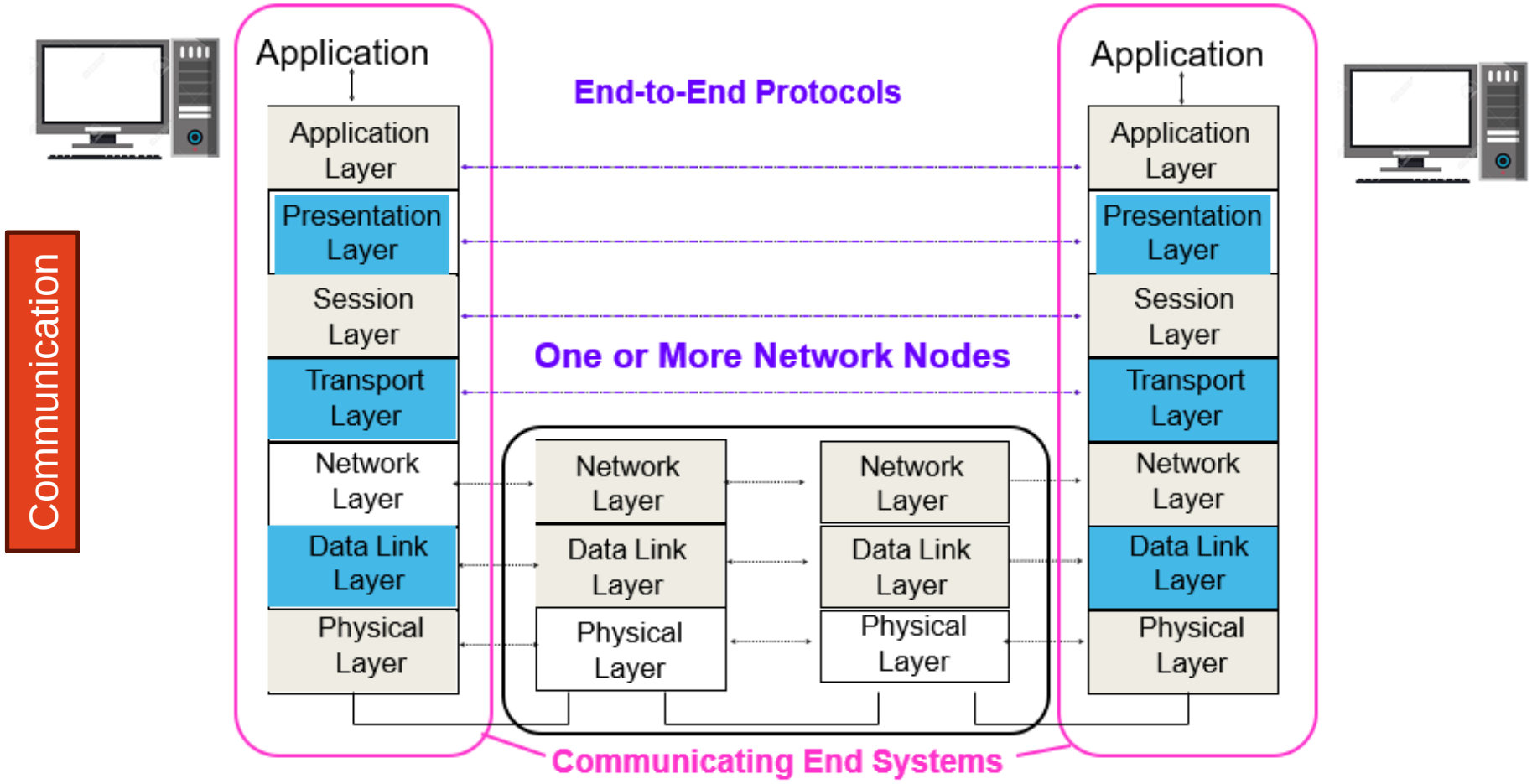
- Error Detection and Correction, etc.

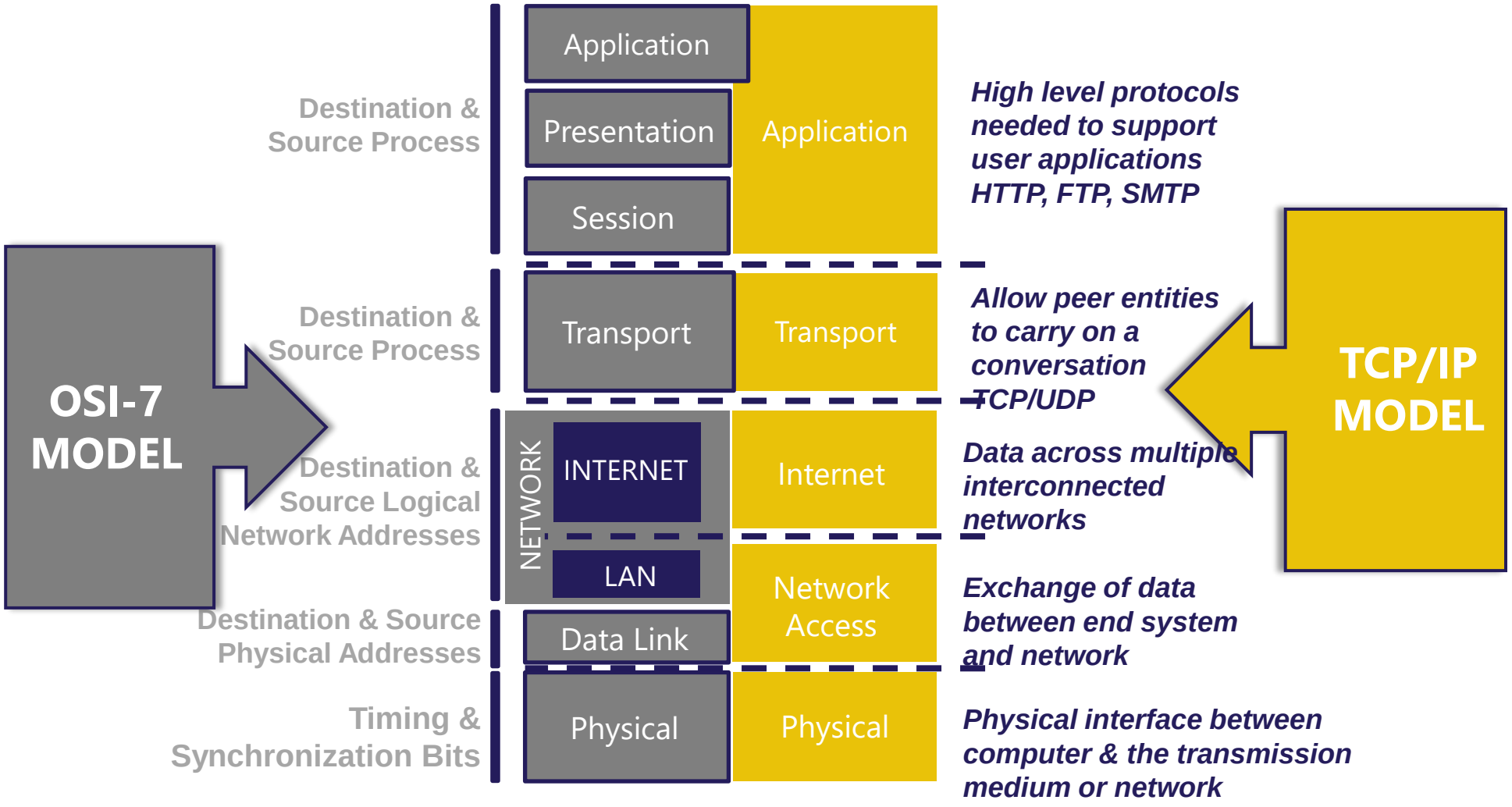
-**Media Access Control (MAC)**

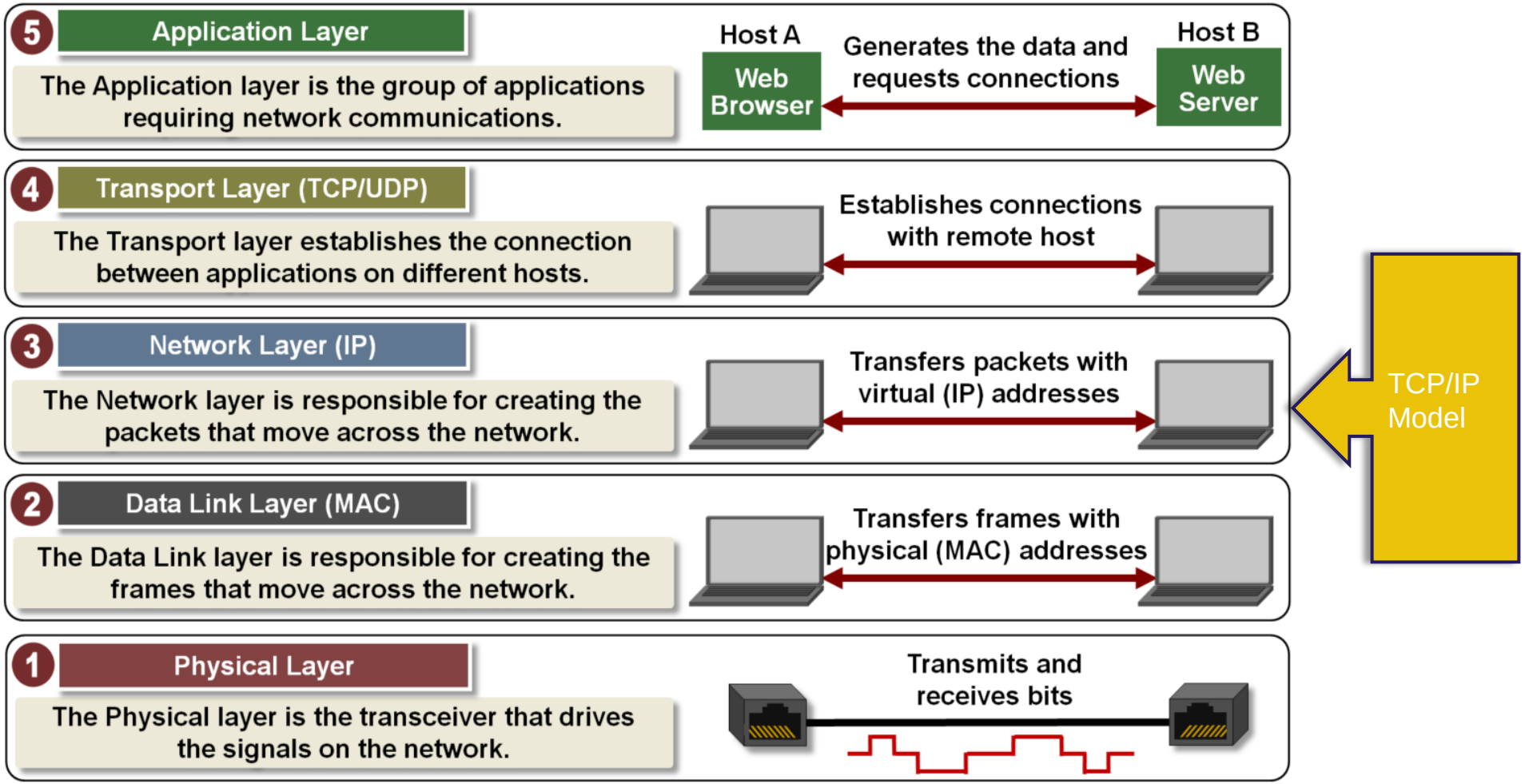
- Framing
- How the media access is controlled and shared

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Segment
Data (Hi !)
Packet
Source IP
Destination IP
Segment
Frame
Source MAC
Destination MAC
Packet
FCS
TRANSPORT LAYER
NETWORK LAYER
DATA LINK LAYER
APPLICATION LAYER
10010111011011001011101101110 11
BITs
BITs
+5V
0V
SIGNALS
AIR
MEDIA
SIGNALS
Hi !
Physical Layer


Communication
Application
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer
End-to-End Protocols
One or More Network Nodes
Network Layer
Data Link Layer
Physical Layer
Network Layer
Data Link Layer
Physical Layer
Application
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer
Communicating End Systems


OSI-7 MODEL
Application
Presentation
Session
Application
Transport
Transport
NETWORK
INTERNET
LAN
Internet
Network Access
Data Link
Physical
Physical
Destination & Source Process
Destination & Source Process
Destination & Source Logical Network Addresses
Destination & Source Physical Addresses
Timing & Synchronization Bits
High level protocols needed to support user applications HTTP, FTP, SMTP
Allow peer entities to carry on a conversation TCP/UDP
Data across multiple interconnected networks
Exchange of data between end system and network
Physical interface between computer & the transmission medium or network
TCP/IP MODEL


5
Application Layer
The Application layer is the group of applications requiring network communications.
Host A
Web Browser
Generates the data and requests connections
Host B
Web Server
4
Transport Layer (TCP/UDP)
The Transport layer establishes the connection between applications on different hosts.
Establishes connections with remote host
3
Network Layer (IP)
The Network layer is responsible for creating the packets that move across the network.
Transfers packets with virtual (IP) addresses
2
Data Link Layer (MAC)
The Data Link layer is responsible for creating the frames that move across the network.
Transfers frames with physical (MAC) addresses
1
Physical Layer
The Physical layer is the transceiver that drives the signals on the network.
Transmits and receives bits
TCP/IP Model

Thank you

